

- 5 (a) (i) Charged particles moving perpendicular to a magnetic field will experience a resultant magnetic force perpendicular to its motion. B1
Hence no work is done. By Newton's 2nd Law, the acceleration of the particles is in the same direction as the resultant force. The direction of the particles changes but not its speed. By Newton's 1st law, upon exit, the particles will move in a straight line with a speed of 4500 ms⁻¹. B1

- (ii) Magnetic force provides centripetal force for particle's circular motion B1

$$F = Bqv = \frac{mv^2}{r}$$

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$$r = \frac{mv}{Bq}$$

C1

$$= \frac{(2.66 \times 10^{-26})(4500)}{(2 \times 10^{-3})(1.6 \times 10^{-19})}$$

A1

$$= 0.374 \text{ m}$$

(b)

$$r = \frac{mv}{Bq} = \frac{P}{Bq}$$

M1

$$P = rBq = (0.2)(2 \times 10^{-3})(1.6 \times 10^{-19})$$

$$= 6.4 \times 10^{-23} \text{ kg m s}^{-1}$$

A1